

# Propagation-Based Degeneracies in Cosmological Redshift Measurements

---

## Abstract

Cosmological redshift is conventionally interpreted as direct evidence of metric expansion of spacetime. However, this interpretation is not observationally unique under current constraints on coherence, surface brightness, and spectral fidelity. In this paper, we analyze a class of propagation-based attenuation frameworks in which redshift arises from cumulative signal loss during transmission, rather than from recessional kinematics. We show that, provided appropriate structural balance conditions are satisfied, attenuation can occur without dispersion, preserving image sharpness and spectral coherence. This induces a degeneracy at the level of observations between expansion-based and propagation-based interpretations. We argue that such degeneracies are not pathological but reflect implicit structural assumptions in cosmological inference. The results provide a neutral baseline for interpreting redshift-related observables, including time dilation and Tolman surface brightness tests, without committing to a specific dynamical cosmology.

## 1. Introduction

The empirical discovery of cosmological redshift remains one of the foundational observations of modern cosmology. Within the standard  $\Lambda$ CDM framework, redshift is interpreted as a consequence of metric expansion, with wavelength stretching directly encoding recessional velocity through the Hubble–Lemaître relation. This interpretation has proven remarkably successful when embedded within a broader dynamical model incorporating dark matter, dark energy, and early-universe inflation.

Despite its success, it is well known that redshift itself is not a direct measurement of expansion. Rather, it is an observed frequency shift that requires theoretical interpretation. Historically, alternative interpretations—including so-called “tired light” models—have been proposed and largely discarded due to conflicts with observations such as image blurring, spectral incoherence, and time dilation. Nevertheless, these rejections typically presuppose specific mechanisms of energy loss that introduce dispersion or stochastic scattering.

In this work, we revisit the question of redshift interpretation from a structural perspective. Rather than proposing a new cosmological model, we ask a narrower question: under what conditions are redshift observations degenerate between expansion-based and propagation-based explanations? By focusing on coherence-preserving attenuation mechanisms, we identify a class of models that evade the classical objections to non-expansion redshift while remaining observationally conservative.

## 2. Observational Status of Redshift Measurements

Redshift is inferred from the displacement of spectral lines relative to laboratory standards. At low redshift, the linear Hubble relation establishes a tight empirical correlation between distance indicators and redshift. At higher redshift, additional observables—including angular diameter distance, luminosity distance, surface brightness, and temporal stretching of light curves—are used to discriminate among cosmological models.

Crucially, most of these observables are indirect. Luminosity distances depend on assumptions about photon conservation, geometry, and propagation. Time dilation measurements rely on source modeling and population statistics. Spectral coherence and image sharpness impose strong constraints on any viable interpretation but do not, by themselves, uniquely select metric expansion.

This leaves open the logical possibility that multiple theoretical mechanisms could map onto the same observational data, provided they satisfy the same structural constraints.

## 3. Classical Propagation-Based Models and Their Limitations

Early tired-light models posited that photons lose energy through interactions with intervening matter or fields. These models typically predict:

- stochastic energy loss,
- wavelength-dependent dispersion,
- angular blurring,
- and violation of observed time dilation relations.

As a result, they are incompatible with high-resolution imaging of distant galaxies and the preservation of blackbody spectra in the cosmic microwave background.

The failure of these models, however, does not constitute a general no-go theorem against all propagation-based explanations. Rather, it rules out mechanisms that lack structural balance and coherence preservation.

## 4. Coherence-Preserving Attenuation Frameworks

We consider a generic class of propagation-based models characterized by the following features:

1. Energy attenuation proportional to path length.
2. Absence of stochastic scattering.
3. Preservation of phase relationships across the wavefront.

Under these conditions, frequency shifts can accumulate deterministically without inducing spatial or spectral decoherence. In classical electrodynamics, analogous behavior is well known in transmission lines satisfying balance conditions that eliminate dispersion.

While we do not commit to a specific microphysical realization, the existence of such frameworks is sufficient to establish observational degeneracy at the phenomenological level.

## 5. Structural Balance and Dispersion Avoidance

A key requirement for coherence preservation is structural balance between dissipative and reactive components of the propagation medium. In engineering contexts, this balance is formalized by conditions ensuring frequency-independent phase velocity.

Translated to a cosmological setting, the implication is that attenuation alone does not imply blurring or spectral distortion, provided the medium—or effective propagation structure—satisfies appropriate balance constraints. This observation undermines the common assumption that any non-expansion redshift must necessarily degrade image quality.

## 6. Degeneracy with Expansion-Based Interpretations

If attenuation scales exponentially with distance while preserving coherence, the resulting redshift–distance relation can mimic the linear Hubble law at low redshift and deviate only subtly at higher redshift. Without independent access to the underlying propagation mechanism, observations alone cannot uniquely distinguish between expansion and attenuation.

This degeneracy is not merely philosophical; it reflects the fact that cosmological inference necessarily relies on structural assumptions beyond raw data.

## 7. Implications for Time Dilation and Surface Brightness Tests

Time dilation is often cited as decisive evidence for expansion. However, if signal attenuation uniformly rescales temporal as well as spectral features, apparent time dilation may arise without invoking metric expansion. Similarly, surface brightness tests constrain but do not eliminate coherence-preserving attenuation scenarios.

The key point is not that expansion-based interpretations are incorrect, but that they are not logically forced by the data alone.

## 8. Discussion: Structural Assumptions in Cosmological Inference

All cosmological models embed implicit assumptions about what constitutes an admissible explanation. Expansion-based models assume that spacetime geometry absorbs the explanatory burden. Propagation-based models instead attribute structure to the transmission process itself.

From this perspective, the debate is not between data and speculation, but between alternative loci of structural commitment.

## 9. Limitations and Open Questions

This work does not provide a microphysical model of attenuation, nor does it attempt to fit detailed cosmic microwave background anisotropies or baryon acoustic oscillations. These remain open challenges for any propagation-based framework. Our aim is narrower: to clarify the logical space of interpretation and identify where observational uniqueness genuinely applies.

## 10. Conclusion

We have shown that cosmological redshift observations admit a structural degeneracy between expansion-based and coherence-preserving propagation-based interpretations. Recognizing this degeneracy does not invalidate the standard cosmological model, but it highlights the role of implicit assumptions in cosmological inference. Propagation-based frameworks, when properly constrained, provide a useful baseline against which claims of observational necessity can be evaluated.

---

*This paper is intended as a conceptual and phenomenological analysis. It does not advocate a specific alternative cosmology but seeks to clarify the structural assumptions underlying redshift interpretation.*